MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2021-22) -- SEMESTER 1

COURSE: PROGRAMMING FOR PROBLEM SOLVING (C LANGUAGE) (BCS-101)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: BCS-101

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) Differentiate between compiler and interpreter with examples.

Q1. (b) Explain the significance of operator precedence and associativity.

Q1. (c) What is a dangling pointer? How to prevent it?

Q1. (d) Write difference between pass-by-value and pass-by-reference.

Q1. (e) What is the output of printf('%d', sizeof(5.5)) in standard C?

Q1. (f) Define static and extern storage classes with scope and lifetime.

Q1. (g) Explain the difference between while and do-while loops.

Q1. (h) How is a 2D array stored in memory using row-major order?

Q1. (i) What are bitwise shift operators (<< and >>)? Give an example.

Q1. (j) Differentiate between malloc() and calloc() dynamic memory allocation.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Write a C program to multiply two matrices of order M x N and N x P with dimension check.

Q2. (b) Explain recursion with a program to solve Tower of Hanoi problem for N disks.

Q2. (c) Develop an algorithm and C function to reverse a singly linked list in-place.

Q2. (d) Explain file pointers and write a program to count lines, words, and characters in a text file.

Q2. (e) Illustrate dynamic memory allocation functions (malloc, calloc, realloc, free) with examples.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain the compilation stages in GCC (Preprocessing, Compilation, Assembly, Linking). [10 Marks]

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(b) Write a complete program to implement Binary Search with time complexity derivation. [10 Marks]

Q4. (a) Differentiate between structure and union with memory layout diagrams. [10 Marks]

-- OR --

(b) Write a program using pointers to find the length of a string without using strlen(). [10 Marks]

Q5. (a) Explain call by reference using pointers with a function to swap two float values. [10 Marks]

-- OR --

(b) Describe dynamic memory reallocation using realloc() with boundary condition handling. [10 Marks]

Q6. (a) Write a program to demonstrate file handling functions fopen, fread, fwrite, and fclose. [10 Marks]

-- OR --

(b) Explain storage classes (auto, static, extern, register) with memory layout and scope. [10 Marks]

Q7. (a) Write a recursive function to compute the N-th Fibonacci number with trace tree. [10 Marks]

-- OR --

(b) Explain macro expansion versus inline functions with syntax and code demonstration. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****